

Lemma 1. *Any closed subset of a compact space is compact.*

Proof. Consider any open cover $\{U_\alpha\}$ of C . Since C is closed, $K \setminus C$ is open. So then $\{U_\alpha\} \cup \{K \setminus C\}$ must be an open cover for all of K . Since K is compact, there exists some finite subcover of $\{U_\alpha\} \cup \{K \setminus C\}$ which covers all of K . Well, $C \subseteq K$, so this finite subcover must cover C as well.

Note that $K \setminus C$ cannot cover $C \subseteq K$ and therefore can't cover all of K . So some subset of $\{U_\alpha\}$ must be included in this subcover. Let $\{U_{\alpha_i}\}_{i=1}^n$ be this subset. If $X \setminus C$ is in this subcover, notice that $X \setminus C$ is disjoint from C , and so $\{U_{\alpha_i}\}_{i=1}^n$ must cover C . Otherwise we once again have that $\{U_{\alpha_i}\}_{i=1}^n$ covers K and therefore C .

So any open cover $\{U_\alpha\}$ of C has a finite subcover, and therefore C is compact. □

Lemma 2. *The continuous image of a compact set is compact.*

Proof. Let K be a compact set in X and $f : K \rightarrow Y$ a continuous map. Now consider any open cover $\{U_\alpha\}$ of $f(K)$. Since f is continuous, each preimage $f^{-1}(U_\alpha)$ is open in K . Now, $\forall x \in K, f(x) \in U_\alpha$ for some α . Therefore, $x \in f^{-1}(U_\alpha)$, and we see that $\{f^{-1}(U_\alpha)\}$ is an open cover for K . Then since K is compact, there exists a finite subcover $\{f^{-1}(U_{\alpha_i})\}_{i=1}^n \subseteq \{f^{-1}(U_\alpha)\}$.

Finally, $K \subseteq \bigcup_{i=1}^n f^{-1}(U_{\alpha_i}) \implies f(K) \subseteq \bigcup_{i=1}^n f(f^{-1}(U_{\alpha_i})) = \bigcup_{i=1}^n U_{\alpha_i} \implies \{U_{\alpha_i}\}_{i=1}^n \subseteq \{U_\alpha\}$ is a finite subcover. So any open cover $\{U_\alpha\}$ of $f(K)$ has a finite subcover, and therefore $f(K)$ is compact. □

Lemma 3. *Any compact subset of a Hausdorff space is closed.*

Proof. Let H be Hausdorff and $K \subseteq H$ a compact subset. We show that $H \setminus K$ is open.

Since H is Hausdorff, for each $x \in H \setminus K$ and each $k \in K$, we can find disjoint open sets $U_k(x), V_k$ containing x and k , respectively. Now, $\{V_k \mid k \in K\}$ is an open cover of K , and since K is compact there exists a finite subcover $\{V_{k_i}\}_{i=1}^n \subseteq V_k$. Set $U_x = \bigcap_{i=1}^n U_{k_i}(x)$ and recall that $U_{k_i}(x) \ni x$ is disjoint with V_{k_i} for each $i \in [n]$.

So by construction, U_x is disjoint from $\bigcup_{i=1}^n V_{k_i} \supseteq K$ placing it completely inside $H \setminus K$. That is, $U_x \subseteq H \setminus K, \forall x \in H \setminus K$. On the otherhand, $x \in U_x, \forall x \in H \setminus K \implies H \setminus K \subseteq \bigcup_{x \in H \setminus K} U_x$.

Finally, we see that then $H \setminus K = \bigcup_{x \in H \setminus K} U_x$, an arbitrary union of open sets, which implies that $H \setminus K$. Therefore, it's complement K must be closed. □

Theorem 4. *If $f : X \hookrightarrow Y$ is a continuous injection where X is compact and Y is Hausdorff, then $X \cong f(X)$. If f is surjective then f is a homeomorphism.*

Proof. Since f is injective, X is in bijection with its image $f(X)$. So we show $f^{-1} : f(X) \rightarrow X$ is continuous by showing the preimage of every closed set is closed.

Consider any closed subset $C \subseteq X$. Since X is compact, by **Lemma 1** we have that C is also compact. Then, since f is continuous, by **Lemma 2**, the continuous image (and preimage of C via f^{-1}) $f(C)$ of the compact set C is also compact. Finally, by **Lemma 3**, any compact subset of a Hausdorff space is closed. So $f(C) \subseteq f(X) \subseteq Y \implies f(C)$ is closed. So we see the preimage $(f^{-1})^{-1}(C) = f(C)$ of any closed set via f^{-1} is closed, and so f^{-1} is also continuous.

So f is a bicontinuous bijection from X to $f(X)$, or equivalently, f is a homeomorphism from X to its image and $X \cong f(X)$. Finally, if $f(X) = Y$ of course f is a homeomorphism from X to Y .

□

Problem 1. Show that local compactness is invariant under continuous open mappings.

Proof. Let $f : X \rightarrow Y$ be a continuous open map, and X a locally compact space. Consider any $x \in X$. By definition $x \in U \subseteq K$ where U is open and K is compact. Since f is a continuous open map, we also have that $f(U)$ is open and then also the continuous image of a compact set is compact by **Lemma 2**, so $f(K)$ is compact. So in fact the statement is immediate; $f(x) \in f(U) \subseteq f(K) \subseteq Y$ and $f(U)$ is open, $f(K)$ is compact.

□

Problem 2. Let $\{X_\alpha\}$ be an indexed family of non-empty sets. Show that $\prod_\alpha X_\alpha$ is locally compact if and only if each X_α is locally compact and X_α is compact for all but finitely many values of α . (You may assume the Tychonoff Theorem.)

Proof. ($\implies$) Suppose that $X = \prod_\alpha X_\alpha$ is locally compact. Each coordinate projection $\pi_\alpha : X \rightarrow X_\alpha$ is a continuous open surjection. The preimage of any open set $U_\alpha \subseteq X_\alpha$ via π_α must be all of X with points in the α -coordinate belonging to U_α and then of course any open set of X either has X_α or some open set U_α in the α -coordinate so its image via π_α is either U_α or all of X_α . It is trivial the map is a surjection. Problem 1 then gives that every X_α is locally compact.

Fix some $x \in X$. X is locally compact so $x \in U \subseteq K$ where U, K are open and compact, respectively. Well, the α -coordinate of U is X_α for all but finitely many α , and so $\pi_\alpha(U) = X_\alpha$ for all but finitely many α . The map is surjective so $\forall y \in X_\alpha, \exists x \in X$ such that $y = \pi_\alpha(x) \subseteq \pi_\alpha(U) = X_\alpha \subseteq \pi_\alpha(K)$ and we have that $\pi_\alpha(K) = X_\alpha$, for all but finitely many α . As previously established $\pi_\alpha(K)$ is compact so X_α must be compact for all but finitely many α .

($\impliedby$) On the other hand, suppose every X_α is locally compact and that for a finite set F , X_α is compact whenever $\alpha \notin F$. Fix $x = (x_\alpha) \in X$. Once again, $\forall \alpha \in F, \forall x \in X_\alpha$ we have $x \in U_\alpha \subseteq K_\alpha$ which are open and compact, respectively. Now set $U_\alpha = K_\alpha = X_\alpha, \forall \alpha \notin F$. All K_α are compact, so by the Tychonoff Theorem, $K = \prod_\alpha K_\alpha$ is compact. Also $x_\alpha \in U_\alpha \subseteq K_\alpha, \forall \alpha$. So then $x = (x_\alpha) \in U = \prod_\alpha U_\alpha \subseteq K = \prod_\alpha K_\alpha$ and U is open. So we see X is locally compact.

□

Problem 3. Let $\{(X_i, \tau_i)\}_{i \in I}$ be a family of topological spaces and define the box topology on $\prod_{i \in I} X_i$ as the topology generated by the set

$$\left\{ \prod_{i \in I} U_i \mid U_i \in \tau_i \right\}.$$

- (a) Show that the product of compact topological spaces, endowed with the box topology, is not necessarily compact.
- (b) Let X be a topological space. We endow $X^{\mathbb{N}} = \prod_{i \in \mathbb{N}} X$ with the box topology. Is the diagonal map

$$\Delta : X \longrightarrow X^{\mathbb{N}} \text{ defined by } x \longmapsto (x, x, x, \dots)$$

continuous?

Here I set $\mathbb{N} = \mathbb{Z}^+$.

Proof. (a) Consider $\{X_i = \{0, 1\}\}_{i \in \mathbb{N}}$ where each copy of $\{0, 1\}$ has the discrete topology. Recall that singletons are open in each copy and in fact form a basis. In the box topology on $X = \prod_{i \in \mathbb{N}} X_i$, every singleton is open, since for all $x = (x_i)_{i \in \mathbb{N}} \in X$,

$$\{x\} = \prod_{i \in \mathbb{N}} \{x_i\}, \text{ a product of open sets in each coordinate.}$$

So X inherits the discrete topology, and an open cover for the space is $\{\{x\} \mid x \in X\}$. Well, the space is infinite so no finite subset of singletons can cover the whole space. So we have found an open cover X without a finite subcover and so this box product of compact topological spaces is not compact.

(b) Not in general. Consider $\mathbb{R}$ (with the standard topology) and $\mathbb{R}^{\mathbb{N}}$. $B = \prod_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = (-1, 1) \cup \left(-\frac{1}{2}, \frac{1}{2}\right) \cup \left(-\frac{1}{3}, \frac{1}{3}\right) \cdots$ is open in $\mathbb{R}^{\mathbb{N}}$. Now consider $\Delta : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}$. If $\Delta(x) = (x)_{n \in \mathbb{N}} \in U = \prod_{n \in \mathbb{N}} U_n$, then $x \in U_n, \forall n \in \mathbb{N}$ by definition. So in fact $\Delta^{-1}(U) = \bigcap_{n \in \mathbb{N}} U_n$. Therefore,

$$\Delta^{-1}(B) = \bigcap_{n \in \mathbb{N}} \left(-\frac{1}{n}, \frac{1}{n}\right) = \{0\},$$

which is not open in $\mathbb{R}$. So $\Delta : \mathbb{R} \rightarrow \mathbb{R}^{\mathbb{N}}$ is not continuous, and we have found a counterexample. That is, the diagonal map is not always continuous.

□

Problem 4. Show that the one-point compactification of $\mathbb{Z}_+$ is homeomorphic to the subspace

$$A = \{0\} \cup \left\{ \frac{1}{n} \mid n \in \mathbb{Z}_+ \right\} \subset \mathbb{R}.$$

Proof. We take denote the one-point compactification $\mathbb{Z}_+^* = \mathbb{Z}_+ \cup \{\infty\}$. Since $\mathbb{Z}_+$ is discrete, its compact subsets are exactly its finite subsets, since its singletons form a basis. So for all distinct $x, y \in \mathbb{Z}_+$, $\{x\} \cap \{y\} = \emptyset$, $x \in \{x\} \subseteq \{x\}$ and lastly $\{\{n\} \mid n \in \mathbb{Z}_+\}$ is a cover of $\mathbb{Z}_+$ without a finite subcover. So $\mathbb{Z}_+$ is Hausdorff, locally compact, but not globally compact. Now by definition, a subset $U \subseteq \mathbb{Z}_+^*$ is open if

U is open in $\mathbb{Z}_+$

or

$$U = \{\infty\} \cup \mathbb{Z}_+ \setminus K$$

For some compact set K in $\mathbb{Z}_+$ which must also be closed by **Theorem 1** since $\mathbb{Z}_+$ is Hausdorff. Notice that then every open set containing ∞ must be $\{\infty\} \cup (\mathbb{Z}_+ \setminus F)$ for some finite subset $F \subseteq \mathbb{Z}_+$. Next, define $f : \mathbb{Z}_+^* \rightarrow A$ by

$$f(\infty) = 0, \quad f(n) = \frac{1}{n}.$$

This is obviously a bijection. Next, consider any open set U in A . It must be the intersection of A and some arbitrary union of open intervals in $\mathbb{R}$. If U doesn't contain 0, then it must simply by definition be some subset of A which will be $U = \{\frac{1}{n}\}_{n \in I \subseteq \mathbb{Z}_+}$ and then the preimage $f^{-1}(U) = \{n\}_I = I$ is an open set in $\mathbb{Z}_+$ and therefore an open set in $\mathbb{Z}_+^*$. Otherwise, $0 \in U \implies U = \{0\} \cup \{\frac{1}{n}\}_{n \in I} \implies f^{-1}(U) = \{\infty\} \cup I$.

Now, since U is open all of its points are interior points and so $0 \in U \implies \exists \varepsilon > 0$ such that $B_\varepsilon(0) \cap A \subseteq U$. Now, pick some $N \in \mathbb{Z}_+$ such that $0 < \frac{1}{N} < \varepsilon$. (which exists by the *Archimedean Property*) For any $n \geq N$, $0 < \frac{1}{n} \leq \frac{1}{N} < \varepsilon$. So $n \in I, \forall n \geq N$. So then any $m \notin I$ must be such that $0 < \frac{1}{N} < \frac{1}{m}$. That is, $0 < m < N \implies F = \mathbb{Z}_+ \setminus I \subseteq [N-1]$. Then, we have that F is finite and therefore compact in $\mathbb{Z}_+$ and by definition $f^{-1}(U) = \{\infty\} \cup I = \{\infty\} \cup (\mathbb{Z}_+ \setminus F)$ where F is compact in $\mathbb{Z}_+$. So $f^{-1}(U)$ is open and f is continuous.

Now, consider any $\frac{1}{n} \in A \setminus \{0\}$ for $n > 1$. The two closest points in $A \setminus \{0\} = \{\frac{1}{n}\}_{n \in \mathbb{Z}_+}$ to $\frac{1}{n}$ via the metric in $\mathbb{R}$ are $\frac{1}{n+1}$ and $\frac{1}{n-1}$. Set $0 < \varepsilon < \min\{|\frac{1}{n} - \frac{1}{n+1}|, |\frac{1}{n} - \frac{1}{n-1}|\}$ for $n > 1$, otherwise, for $n = 1$, set $0 < \varepsilon < 1/2$. Notice that $B_\varepsilon(\frac{1}{n}) = (\frac{1}{n} - \varepsilon, \frac{1}{n} + \varepsilon)$ is open in $\mathbb{R}$ and in fact since it excludes the closest other points to $\frac{1}{n}$ in A , $\forall n \in \mathbb{Z}_+$. So we have that $B_\varepsilon(\frac{1}{n}) \cap A = \{\frac{1}{n}\}$ is open for all $\frac{1}{n} \in A \setminus \{0\}$. So non-zero singletons are open in A .

Therefore, for any distinct $\frac{1}{m}, \frac{1}{n} \in A \setminus \{0\}$: $\{\frac{1}{m}\}$ and $\{\frac{1}{n}\}$ are disjoint open sets, and otherwise for distinct $a, b \in A$ without loss of generality $a = 0, b = \frac{1}{n}$ and so pick $0 < \varepsilon_m < \frac{1}{m}$ and we see that $B_{\varepsilon_m}(0) \cap A = \{0\} \cup \{\frac{1}{i}\}_{i > m}$ and $\{\frac{1}{m}\}$ are disjoint open sets. So then A is Hausdorff, and we already have that $\mathbb{Z}_+^*$ is compact. Therefore, f is a continuous bijection from a compact space to a Hausdorff space, and so finally by **Theorem 4**, we have that f is a Homeomorphism; $\mathbb{Z}_+^* \cong A = \{0\} \cup \{\frac{1}{n}\}_{n \in \mathbb{Z}_+} \subseteq \mathbb{R}$.

□

Problem 5. Let X denote the rational points of the interval $[0, 1] \times \{0\} \subset \mathbb{R}^2$. Let T denote the union of all line segments joining the point $p = (0, 1)$ to the points of X .

- (a) Show that T is path connected, but it is locally connected only at the point p .
- (b) Find a subset of $\mathbb{R}^2$ that is path connected but is locally connected at none of its points.

Proof. (a) Recall the definition of path connectedness: A space X is path connected if $\forall x, y \in X$, there exists a continuous map $f : [0, 1] \rightarrow X$ such that $f(0) = x$, $f(1) = y$. Now consider any $a, b \in T$ and define $f : [0, 1] \rightarrow T$ as the map that (in this order) sends all points along $[0, 1/2]$ to the points along the segment from a to p and then all the points from $[1/2, 1]$ to the points along the segment from p to b .

Now, consider any open set in T . It is the intersection of some arbitrary union of open balls in $\mathbb{R}^2$ and T . The intersection of an open ball and T will just be an arbitrary union of 'subsegments' of T contained in that open ball. Well, the only points in T with a preimage via f lie on the segment from a to p or the one from p to b . That is, the preimage of any subsegment of a segment other than $[a \rightarrow p]$ or $[p \rightarrow b]$ is simply $\emptyset$. So pulling back from any open set U of T via f , will give us the union of preimages of its subsegment members, which we just established will only give us the union of preimages of subsegments of $[a \rightarrow p]$ and $[p \rightarrow b]$ contained in U . Well, the preimages of these are open intervals in $[0, 1]$, half open intervals in $[0, 1]$, or the whole space, which are all open in $[0, 1]$. (We can't form a closed paths using arbitrary unions of open sets in $\mathbb{R}^2$, otherwise we could do so in $\mathbb{R}$) If no portion of either segment exists in our open set U then its preimage is $\emptyset$ which is open. So f is open for any $x, y \in T$ and T is path connected.

Consider any open neighborhood U of p in $\mathbb{R}^2$, it must contain some open ball $B_\varepsilon(p) \subseteq U$ for some $\varepsilon > 0$. Well, just use the same map and argument as before for any distinct $a, b \in B_\varepsilon(p) \cap T = U_\varepsilon \subseteq U$ to see that U_ε is a path connected, and therefore connected open neighborhood of p contained in U satisfying local connectivity at p ; we can always draw a path from $f(0) = a$ back to $f(1/2) = p$ and then a path from $f(1/2) = p$ to $f(1) = b$.

Suppose T is locally connected at any $x \in T \setminus \{p\}$, and choose $\varepsilon > 0$ small enough that $p \notin B_\varepsilon(x) = U$. Since T is locally connected at x there exists a connected open neighborhood $V \subseteq U$ of x . Now, since V is open, it contains some open neighborhood $B_\delta(x)$ for some $\delta > 0$ and we have some $0 < 1/M < \delta$ and so a subsegment of the spokes from p to $(\frac{1}{M}, 0)$ and from p to $(\frac{1}{M+1}, 0)$ belong to it. Therefore, choose some irrational $\frac{1}{M+1} < c < \frac{1}{M}$ whose spoke doesn't belong to T . The open planes $H_L(c)$ and $H_R(c)$ on either side of the line ℓ_c this spoke lies on are open in $\mathbb{R}^2$ are open and cover V since they cover $\mathbb{R}^2$ with the ℓ_c removed, which isn't in V . So when intersected with T and then V , we get a decomposition of V into two disjoint open neighborhoods; $V = ((H_L(c) \cap T) \cap V) \cup ((H_R(c) \cap T) \cap V)$, which contradicts that V is connected. So we see T is locally connected at p and nowhere else.

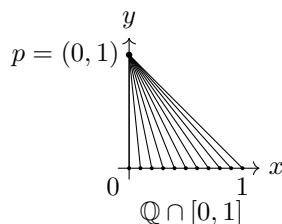


Figure 1: The space T , consisting of the line segments from $p = (0, 1)$ to $(q, 0)$ for $q \in \mathbb{Q} \cap [0, 1]$.

(b) Now let $T_1 = T$ with hub $p_1 = (0, 1)$ and T_2 be the set of all segments from $p_2 = (0, 0)$ to rational points along $X' = ([-1, 0] \times 1) \subseteq \mathbb{R}^2$. Slide T_2 up 1 and reflect about $y = 1, x = 0$, consecutively, to see that $T_2 \cong T_1 = T$. Now consider $Y = T_1 \cup T_2$, a union of two copies T_1, T_2 of T joined at their hubs p_1, p_2 by a 'spine' subsegment of the y -axis. In fact, to be clear: the spine segment $0 \times [0, 1]$, which includes both hubs, belongs to both T_1 and T_2 .

Suppose Y is locally connected at any non-hub point x that comes from a copy of T , and choose $\varepsilon > 0$ small enough that $U = B_\varepsilon(x) \cap Y$ is disjoint from the other copy of T and the spine $\{0\} \times [0, 1]$. (which includes p_1, p_2) U is contained completely in this copy of T and we have a connected open neighborhood $V \subseteq U$ containing some ball $B_\delta(x)$. Then we make the same argument (using halfplanes bounded by an irrational spoke excluded from this copy of T) as we did in (a) and get a contradiction proving that Y is not locally connected at any non-hub point in either copy of T .

Now suppose Y is locally connected at any non-hub point along the spine; any $x \in \{0\} \times (0, 1) \subset Y$. Choose $\varepsilon > 0$ small enough that the open neighborhood $U = B_\varepsilon(x) \cap Y$ doesn't contain p_1 or p_2 . It must have a connected open subneighborhood $V \subseteq U$ of x and then $\exists \delta > 0$ small enough that $B_\delta(x)$ is contained in V . By the *Archimedian Property* there exists a rational point $0 < \frac{1}{N} < \delta$ whose spoke to p_1 has a subsegment contained in $B_\delta(x)$. Now pick some irrational $0 < c < \frac{1}{N}$, its spoke to p_1 is not contained in V by definition and so it partitions V making same halfplane argument as before, contradicting that V is connected. So Y is not locally connected at any point outside of $\{p_1, p_2\}$ now.

Finally, suppose Y is locally connected at a hub p_i in either copy of T within Y , and pick $\varepsilon > 0$ small enough that $U = B_\varepsilon(p_i) \cap Y$ doesn't contain the other hub.

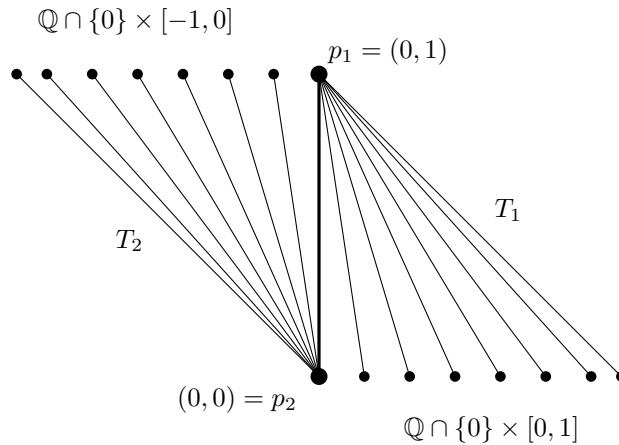


Figure 2: The space $Y = T_1 \cup T_2 \cup (\{0\} \times [0, 1])$.

□

Problem 6. Define the cone CX to be the quotient space obtained by identifying $X \times \{1\}$ in $X \times [0, 1]$ to a point. Prove that CX is path connected. Deduce that any X is a subspace of a path connected space.

Proof. Let $q: X \times [0, 1] \rightarrow CX$ be the quotient map, and let $v = q(X \times \{1\})$ be the cone point. For any $q(x, t) \in CX$, the map

$$\gamma_{x,t}(s) = q(x, (1-s)t + s), \quad 0 \leq s \leq 1,$$

is a path from $q(x, t)$ to v . Any two points can therefore be joined by following one such path to v and then following the reverse of another. Thus CX is path connected.

The set $X \times [0, 1)$ is open and saturated for the quotient map, and no two of its points are identified. Hence the restriction

$$q|_{X \times [0, 1)}: X \times [0, 1) \longrightarrow CX \setminus \{v\}$$

is a homeomorphism. In particular, $x \mapsto q(x, 0)$ embeds X as the subspace $q(X \times \{0\})$ of the path connected space CX . $\square$